

UNIVERSITY OF GHANA
COLLEGE OF BASIC AND APPLIED SCIENCE



DCIT 316: COMPUTATIONAL MODEL FOR SOCIAL MEDIA MINING

SEMESTER PROJECT PROPOSAL

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DEPARTMENT OF COMPUTER SCIENCE

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Project Title

Tokenization Efficiency in Ghanaian Low-Resource Languages: A Comparative Evaluation of Modern Multilingual Tokenizers for Twi and Ewe

Team Declaration

Team status: Solo

Role	Full name	Student ID	GitHub username	Email
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Proposed team name: GhanaToken

Problem Definition

Modern NLP systems depend on tokenizers to convert text into units that language models process. Because multilingual vocabularies are often shaped by high-resource languages, low-resource languages may be split into many subword tokens. Recent African NLP research shows that high token fertility can create computational and performance disadvantages.

This project investigates this problem specifically for two Ghanaian languages: Twi and Ewe. It will compare selected modern multilingual tokenizers on the same Ghanaian-language data and then test whether a small vocabulary trained on Twi/Ewe text can reduce unnecessary subword fragmentation.

The study will measure token fertility, tokens per word, tokens per character, word-fragmentation rate and relative tokenization efficiency. A small downstream Twi social-media sentiment experiment may be included only if the core tokenizer study is completed early.

Motivation

Recent work has established that tokenizer design can create systematic differences between languages. The African Language Tax reports measurable cost, latency and context penalties for African languages across 11 frontier and open tokenizers, while The Token Tax reports that higher token fertility predicts lower accuracy across 16 African languages. These studies establish the broader African problem, but they do not provide a focused two-language Ghanaian experiment using the same controlled dataset and a Ghanaian-specific vocabulary construction.

The Ghanaian NLP Landscape identifies Ghanaian languages as an important but under-resourced area of NLP, while GhanaNLP has recently released parallel corpora for Twi, Fante, Ewe, Ga and Kusaal. This makes a focused Twi/Ewe tokenizer study feasible without scraping or collecting new personal data.

Twi and Ewe are selected to keep the study realistic for one student working within a short period of time while still providing a meaningful cross-language comparison. The project is therefore positioned as a focused Ghanaian case study and reproducible benchmark, not as a claim that tokenization research itself is new.

Research Question

How efficiently do selected modern multilingual tokenizers represent Twi and Ewe, and can a vocabulary train on Ghanaian-language text reduce token fragmentation compared with existing multilingual tokenizers?

Secondary question: Does improved tokenization efficiency correspond to improved performance on a small Twi social-media sentiment-classification task?

Research Objectives

- Measure tokenization efficiency for Twi and Ewe using selected modern multilingual tokenizers.
- Compare token fertility and word fragmentation between Twi and Ewe.
- Train a lightweight Ghanaian-language BPE or SentencePiece tokenizer using Twi/Ewe data.
- Determine whether the Ghanaian-specific vocabulary reduces token fragmentation relative to the selected multilingual baselines.
- Produce a reproducible dataset-processing and tokenizer-evaluation pipeline.
- Test whether tokenizer differences are reflected in Twi sentiment classification.

Data Source and Access Method

Field	Details
Exact data source	GhanaNLP Parallel Corpora (Twi and Ewe portions); AfriSenti benchmark (Twi subset, optional)
Access method	Public research datasets (no Twitter/X scraping or private-data collection)
Link/reference	GhanaNLP Parallel Corpora; AfriSenti sentiment dataset
License/access condition	Open/Public research use
Unit of analysis	Sentence / Sentence pair
Estimated sample size	Balanced sample of ~2,000–5,000 sentences per language for initial experiments
Can raw data be shared in GitHub?	Public research datasets (follow source repository redistribution terms)

Ethics and Privacy

The project will use existing public datasets and will not collect private information. No attempt will be made to identify social-media users or recover removed information. Results will be reported in aggregate. Raw social-media content will not be redistributed unless the dataset licence explicitly permits it; the repository will prioritize code, dataset references and derived statistics.

Method / Pipeline

1. Load and inspect Twi and Ewe corpora.
2. Apply consistent Unicode normalization and minimal preprocessing.
3. Create balanced evaluation samples.
4. Run the same text through three representative multilingual tokenizers.
5. Calculate token fertility, tokens/word, tokens/character and fragmentation.
6. Train a lightweight BPE/SentencePiece tokenizer on the Ghanaian-language training data.
7. Evaluate the Ghanaian tokenizer on a held-out test split.
8. Compare it with the existing tokenizers.
9. If time permits, run a small Twi AfriSenti sentiment baseline.
10. Conduct error analysis and write conclusions.

Evaluation Plan

Metrics: Token fertility (subword tokens per whitespace-separated word), Average tokens per sentence, Tokens per character, Percentage of words split into multiple tokens, Vocabulary coverage, Compression ratio, Accuracy, Macro F1, Precision, Recall, Confusion matrix, and Qualitative error analysis.

Explanation: Primary tokenizer efficiency and fragmentation metrics evaluate subword tokenization behavior across languages, while standard classification metrics (Accuracy, Macro F1, Precision, Recall, Confusion matrix, and qualitative error analysis) evaluate the downstream sentiment experiment; all models process identical held-out sentences to ensure controlled comparison.

Literature Grounding

Paper citation	Problem studied	Method used	Dataset/source	How it supports our project
Muhammad et al. (2023) AfriSenti	No gold sentiment labels for African languages	Crowdsourced 3-class annotation	110k+ tweets, 14 languages	Provides the optional Twi social-media sentiment dataset for downstream validation
Parallel Tokenizers	Vocabulary design limitations in cross-lingual transfer of low-resource languages	Empirical analysis of cross-lingual vocabulary design	Cross-lingual corpora	Motivates testing whether vocabulary design and language-specific tokenization can improve low-resource representation and fertility balance
The African Language Tax	Tokenization cost, latency, and context penalties in frontier LLMs	Quantitative analysis of tokenization overhead	Frontier LLM benchmarks across African languages	2026 evidence of African-language tokenization cost, latency and context penalties; this project applies the broader question to Twi and Ewe
The Token Tax	Systematic bias and fertility disparities in multilingual tokenizers	Cross-lingual tokenization efficiency analysis	16 African languages	Shows across 16 African languages that higher token fertility is associated with lower accuracy, motivating fertility as a primary metric
The African Languages Lab	Lack of systematic data and language-specific models for African NLP	Collaborative data collection and model development	Multi-language African corpus (including Twi)	Shows current African NLP efforts in systematic data collection and language-specific model development; Twi is among evaluated languages
The Ghanaian NLP Landscape	Resource gaps and limited evaluation in Ghanaian NLP	Landscape review and gap analysis	Ghanaian language resources	Provides Ghana-specific context on the state of Ghanaian NLP and resource gaps
GhanaNLP Parallel Corpora (2026)	Scarcity of high-quality parallel evaluation resources for Ghanaian languages	Professional creation of parallel sentence pairs	Twi, Fante, Ewe, Ga, and Kusaal parallel pairs	Provides the practical Twi/Ewe data resource for a controlled Ghanaian-language experiment
ABENA / GhanaNLP Twi BERT work	Baseline subword tokenization and language modeling performance for Twi	Language model pretraining and tokenizer evaluation	Twi corpus	Establishes prior Twi-specific language modelling/tokenizer work, so this project focuses on comparative modern evaluation rather than claiming to invent Twi tokenization
Disparities in LM Tokenizers	Multilingual tokenizer design creates inherent language-level unfairness	Comparative evaluation of multilingual tokenizers	Multilingual text evaluation sets	Provides earlier evidence that multilingual tokenizer design can create language-level disparities
Do All Languages Cost the Same?	Cost and efficiency imbalances caused by subword tokenization	Token-per-character and cost analysis	Multi-language parallel text	Supports evaluating tokenization efficiency/cost rather than model accuracy alone

Research Gap / Proposed Contribution

The project does not claim that tokenization of Ghanaian languages has never been studied. Twi-specific tokenizer work already exists, and recent African research has established the broader tokenization-tax problem. The proposed gap is narrower: a focused, reproducible comparison of modern multilingual tokenizers on Twi and Ewe under the same evaluation framework, followed by an experiment testing whether a Ghanaian-language-specific vocabulary reduces fragmentation.

The contribution will therefore be empirical evidence and a reusable benchmark pipeline, rather than a claim of inventing a new tokenization algorithm.

Expected Output

The project will produce Python source code, notebooks, a reproducible tokenizer evaluation pipeline, tokenization comparison tables and figures, a trained Twi/Ewe tokenizer, aggregate results, an optional Twi sentiment baseline, error analysis, a data card, a tokenizer/model card where applicable, a technical report, GitHub repository and presentation slides.

The main research result will be a clear comparison of how existing tokenizers represent Twi and Ewe and whether Ghanaian-language vocabulary construction measurably reduces subword fragmentation.

Risks and Fallback Plan

Risk	Possible problem	Fallback
Data access	A corpus version is unavailable or difficult to load.	Use the public GhanaNLP distribution/Hugging Face version or reduce the sample size.
Tokenizer training	Custom tokenizer training fails or gives unstable results.	Complete the existing-tokenizer benchmark and report custom vocabulary training as an additional experiment.
Computational limits	A selected model/tokenizer is too heavy.	Evaluate the tokenizer component directly or replace it with a lighter representative tokenizer.
Optional sentiment task	The downstream experiment takes too long or is unreliable.	Drop it; the tokenizer benchmark remains the complete core project.

Final Declaration

I would like to state that the proposal presented reflects work that I can justify and explain in its entirety. Also, I will uphold the principles of academic research by avoiding the use of scraping, private data, artificial citations, or results without approval, and I will disclose AI assistance honestly and verify AI-assisted work.